KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

UNIT THREE (A)

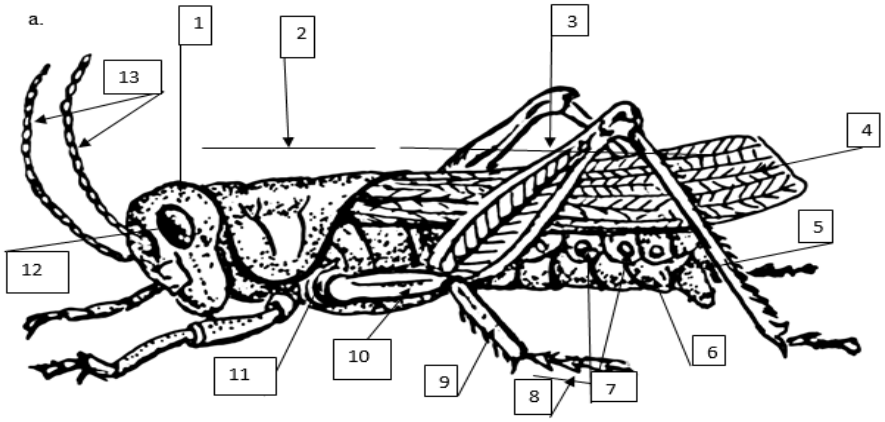
TOPIC: DESCRIBING INSECT ANATOMY AND PHYSIOLOGY

Objectives: by the end of this unit, SSBAT;

- Describe the external parts of an insect
- Explain the functions of external parts

3.1. External parts and the functions of the external parts of insects

An understanding of insect structure is necessary for distinguishing insects from each other and from other animals. It is also important to know how insects function, live and move about in the environment. Insects display a wide diversity in form. Therefore, it is necessary to use a generalised model (the grasshopper) in discussing insect morphology.



Adult insects are characterized by having three **tagmata** (singular tagma) comprising the head (1), thorax (2) and abdomen (3) (three distinct body parts). The grouping of insect body segments into well-defined functional regions or body parts is known as TAGMOSIS.

The head, besides bearing the mouthparts, the insect head contains many sensory organs that perceive and process external information (antennae, eyes and ocelli). The thorax is the middle region of the insect body and bears the locomotory organs (legs and wings).The abdomen is the posterior body part and also consists of segments. It contains most of the organs associated with metabolism and reproduction.

3.1.1. **Exoskeleton**

3.1.1.1. Cuticle

The insect structure forms the body wall and is protected by a hard external "shell" or integument called the **exoskeleton**, which is composed of cuticle that is secreted by the epidermis. Cuticle is formed of chitin (an amino-sugar polysaccharide) and protein, and there are several layers that vary in development and composition on different areas of the body to allow flexibility or strength of the body wall. The exoskeleton is like a spring but does not stretch at certain times during growth.

Functions of the exoskeleton

i. It provides rigidity to the insect body. It's a structure that makes the insect firm.

ii. It limits the size the insect can attain without adjustment. It acts as a spring and it is a flexible cuticle that can expand and accommodate the growth of the insect.

iii. It must shed in order for insects to grow through moulting from one stage to the next stage. Moulting and growth occur mostly during larval or nymphal instars, and adult insects never moult again.

iv. It's a shield for internal organs against external mechanical injuries and desiccation from adverse environmental conditions such as temperature.

v. It acts as a framework for muscle attachment. It supplies the support system for the attachment of the insect muscles.

vi. It's a surface for sensory receptor system which keeps the insect in contact with the environment.

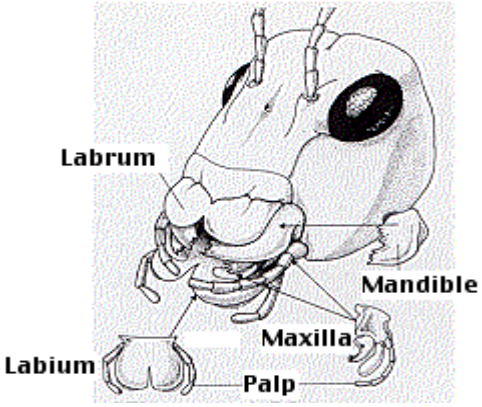
3.1.1.2. Endoskeleton
3.1.1.3. Epidermal cells are single cell layers which secretes a substance that forms up the cuticle.

3.1.2. The head

The head of the insect is made of a cranium which bears the eyes, the antennae and mouth parts. The head is a structure specialised for feeding and for sensing and integrating information about the environment. The insect cranium is a hardened capsule with an opening to the mouth and thorax. It is attached to the thorax by a way of a short neck, or cervix.

The orientation of the cranium varies among different insects, positioned vertically (mouthparts pointed down, hypognathous), horizontally (mouthparts pointing forward, prognathous), or oblique (mouthparts pointed backwards, opisthognathous). Insects like leafhoppers and mosquitoes, which pierce and suck liquid food often, have much more delicate crania.

3.1.2.1. Mouthparts
These are some of the most distinctive features of insects, and their structure tells a great deal about their feeding habits. Mouthparts are greatly varied, but in their primitive form, for example, in grasshoppers and beetles, they include a labrum (upper lip), a pair of chewing mandibles (jaws), a pair of maxillae (second jaws) and a labium (lower lip). These structures surround the mouth and form the pre – oral cavity. In addition, a central tongue like hypopharynx drops from membranous floor of the cranium, behind the mouth, and bears the opening of the salivary ducts. The interior, fleshy surface of the labrum, endowed with numerous sensory structures is referred to as the labrum – epipharynx.



3.1.2.1.1. Mouth types

a. Chewing mouth

This type of mouth enables the insect to bite off pieces of food with the cutting area of the mandible. Both mandibles and maxillae then orient and manipulate the food and,

to a degree, the grinding area of the mandible masticates it. During this activity, saliva is applied, after which the food bolus is guided opening with the aid of the maxillae, labium and hypopharnx.

b. Piercing-sucking mouth

These mouth types are very common among insects. They are modified to pierce the epidermis of the plant or the skin of animals and to suck up sap or blood. In plant feeders such as cicadas and aphids (Homoptera) the piercing-sucking needle is formed from hair like stylets fitted closely together.

In blood-feeding insects like mosquitoes (Diptera: Culicidae), an elongate labium also forms a protective sheath for six rather than four stylets. In feeding, the maxillary and mandibular stylets work together as a needle to penetrate the host.

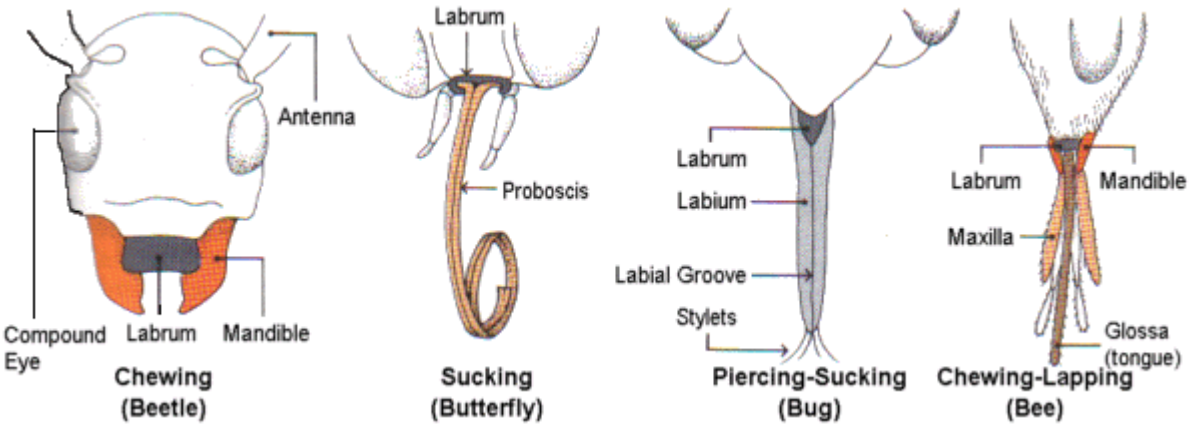
c. Rasping-sucking mouth

These are primitive form of the piercing-sucking type and are found in thrips (Thysanoptera). Thrips mouth parts have a cone shaped beak formed from the clypeus (a plate above the labrum in chewing insects). Thrips use the beak to rasp host tissues and take up liquid food through the stylets.

d. **Siphoning mouths** are a highly specialised type for the uptake of flower nectar and other liquids by butterflies and moths (Lepidoptera). A long proboscis composed of maxillary elements forms tubes through which food passes. The tubes are sponging mouth type and are high specialised structures in house flies *(Musca domestica)* and their close relatives. With this type the mandibles and maxillae are non-functional, and the remaining parts form a proboscis with a fan shaped sponge at the tip. Liquid food is 'mopped up' by the capillary action of this sponge, and if not liquid, salivary secretions through the mouth make it so.

e. Chewing lapping

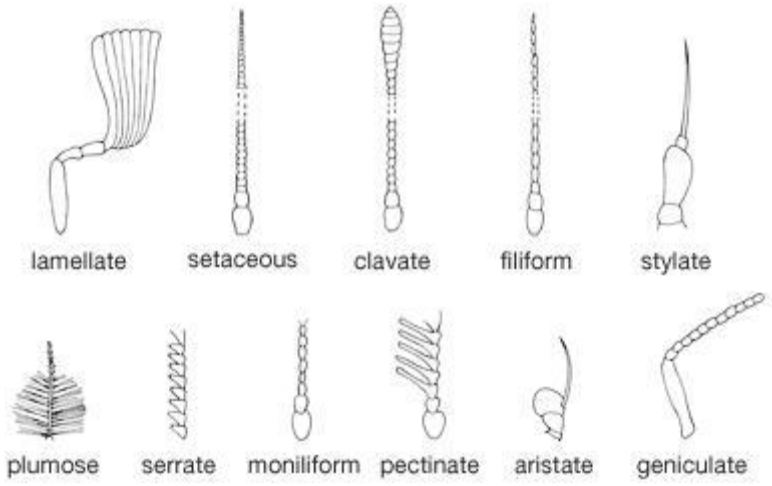
This is the mouth type in bees and wasps (Hymenoptera). With this type the mandibles and the labrum are similar to the chewing type and are used in moulding wax, manipulating nest materials, and grasping prey. The maxillae and labium form a channelled proboscis through which saliva is discharged and nectar is drawn up.



3.1.2.2. Antennae

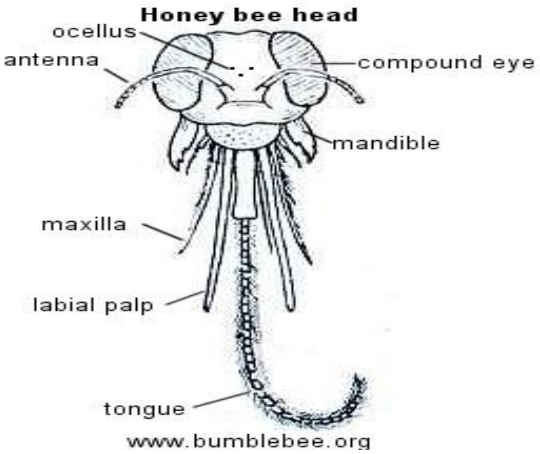
Pair of variously shaped appendages called antennae is found on the head of insects below the eyes. These are movable and contain sensory structures that allow insects to detect odours, vibrations and other environmental stimuli. Antennae shape is strikingly different among insect species and often between sexes of the same species. It is for this reason that antennae are frequently used in insect identification and sex determination.

Major antennal types include: i. Filiform (threadlike), ii. Serrate (saw like), iii. Moniliform (beadlike), iv. Clavate (club-shaped), v. Lamellate (plate like), vi. Pectinate (comb-shaped) and vii. Plumose (featherlike) to mention but a few. Below are some of the images.



3.1.2.3. Compound eyes

Located on each side of the head of most adult insects are prominent compound eyes which consist of many hexagonal elements. These elements are known as **facets.** They number from only a few to as many as 28,000 in dragonflies. In addition to these, most adult insects have several simple eyes, or **Ocelli.** The ocelli are located between compound eyes on the front of head. Most often, they are small and have a simple lens. Many immature insects e.g. caterpillars (Lepidoptera order), have only simple eyes and other insets may have no eyes at all, as in some springtails (Collembola order).



3.1.3. Part of the thorax

The thorax is a raged "box" of three distinct segments, from the back of the head. These segments are **prothorax, mesothorax** and **metathorax.** Each thoracic segment bears a pair of jointed legs, and in most adult insects the mesothorax and metathorax each have a pair of wings. Each thoracic segment is composed of hardened plates that give it rigidity. The upper plate is called notum, and the lower plate is called the sternum. Side plates called Pleura (single pleuron), are present and most well developed on the winged segments.

3.1.4. The abdomen

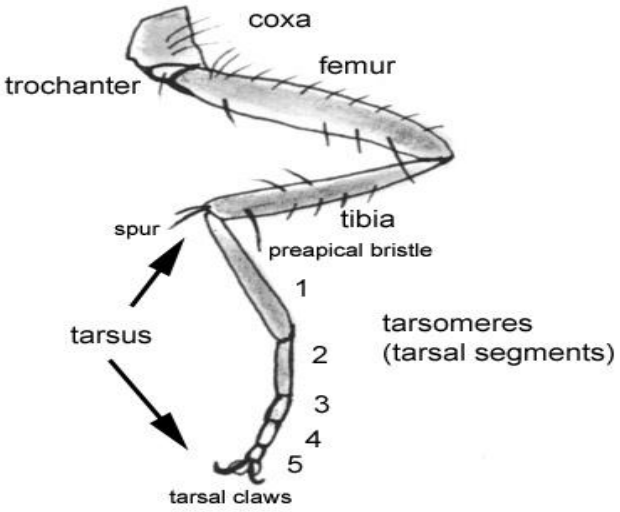
The abdomen is made of several segments which end just where the anus ends. The terminal segment is the sensory organ called the **cerci.** Some insects do have reproductive structures that arise from the last segment to form what is called as **ovipositor** (egg laying device). Although adult insects lack functional legs on the abdomen, many immature insects such as caterpillars have fleshly prolongs that aid in locomotion. Because of their tremendous diversity and distinctiveness, the abdomen is often used to identify insect species.

3.1.5. Parts of an insect leg

Insect legs are articulated appendages comprising five segments from the body outwardly and these are the **coxa, trochanter, femur, tibia** and **tarsus**. The coxa fits into a cuplike depression in the body, allowing multidirectional movement. The trochanter connects the coxa with the femur, usually the largest leg segment. The tibia is often a long, slender structure with downward pointing spines that aid in climbing. The tarsus usually made up of several sub segments called **tarsomere**, terminates in a **pretarsus**. The pretarsus usually includes a pair of claws for adhesion. If a pad is present at the base of each claw, the pads are called **pulvilli**.

Some of the functions that insect legs serve are:

- ✓ Catching their prey.
- ✓ Running.
- ✓ Swimming.
- ✓ Walking or hopping.
- ✓ Digging or burrowing.



3.1.6. The Wings of an insect

Along with birds and bats, insects are the only other animals with wings. Insect wings show much diversity, varying in shape, texture and coloration. Most insects have two pairs, except flies and a few other groups, which have only one pair. In flies each metathoracic wing has been modified to form a knobbed guidance organ called a **halter** (plural **haltere).** Other wing modifications include the tough, protection cover in beetles (Coleoptera) formed from the front wings, the elytra. Wings are used as a sort of road map in analysing and identifying certain insects groups. The most common wing is the membranous one with a system of veins running throughout. These veins give strength and rigidity to the wings. Some insects have no wings. Silverfish (Thysanura) and few other insects probably has wingless ancestors (primitively wingless), whereas others, such as flea and lice, probably evolved from a winged ancestor (secondary wingless). Aphids have non winged forms only at certain times of the year. Except for mayflies (Ephemeroptera), all immature insect lack functional wings.